Year 13 Further Maths - Correlation

Similar questions - Product-moment correlation coefficient tests

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A collector compared the prices of some items when new in 1980 (£x) and the prices of second-hand copies on a website in 2020 (£y).

Item	A	B	C	D	E	F	G	H	I	J	K	L
x	0.80	0.60	0.75	1.10	0.45	1.30	1.60	0.55	1.05	0.30	0.25	0.40
y	5.40	6.50	2.40	6.10	3.50	5.00	7.60	2.20	3.40	9.10	1.50	2.40

$n = 12$, $\sum x = 9.15$, $\sum y = 55.10$, $\sum x^2 = 8.9425$, $\sum y^2 = 316.6100$, $\sum xy = 45.81$.

(a) You are given that Pearson's product-moment correlation coefficient is 0.340.

(i) State what this tells you about a scatter diagram of the data. **[1]**

(ii) Test at the 5% significance level whether there is evidence of positive correlation between the two sets of prices. **[5]**

(b) The collector notices that item J (second-hand price £9.10) is unusually expensive and decides to ignore it. Calculate the value of the correlation coefficient for the other 11 items and carry out the test again at the 5% level. **[4]**

2 The directors of a company believe that there are more computer failures in the head office when temperatures are higher. They record the temperature t°C and the number of failures x on each of 10 random days.

$$n = 10, \quad \sum t = 276, \quad \sum x = 36, \quad \sum t^2 = 7912, \quad \sum x^2 = 154, \quad \sum tx = 1073.$$

(a) State which of the words *dependent, independent, controlled, response* may be applied to t. **[1]**

(b) Calculate the product-moment correlation coefficient. **[2]**

(c) Explain why a 1-tail test is appropriate, and test the directors' belief at the 1% significance level. **[6]**

(d) The directors decide to use temperatures in °F instead, where $F = \frac{9}{5}C + 32$. State the value of the correlation coefficient that would result. **[1]**

3 A newspaper claims that "taller students have heavier backpacks". Maya records the height h cm and backpack mass m kg of 15 random students.

$$n = 15, \quad \sum h = 2584, \quad \sum m = 609, \quad \sum h^2 = 446340, \quad \sum m^2 = 25313, \quad \sum hm = 105712.$$

(a) Calculate the product-moment correlation coefficient. **[2]**

(b) State what your answer tells you about a scatter diagram of the data. **[1]**

(c) Test the newspaper's claim at the 5% significance level. **[5]**

(d) Explain whether the answer to part (a) would be likely to change if backpack mass were measured in grams. **[1]**

Question 1

(a)(i) The points show weak positive correlation (the points are scattered with a slight upward trend)

(ii) H_0: $\rho = 0$; H_1: $\rho > 0$. Critical value ($n = 12$, 5%, 1-tail) 0.4973. $0.340 < 0.4973$, so there is not enough evidence to reject H_0: no evidence of positive correlation

(b) Without J: $r = 0.717$; critical value ($n = 11$) 0.5214. $0.717 > 0.5214$, so reject H_0: there is evidence of positive correlation once the outlier is removed

Question 2

(a) t is the independent variable (and not controlled since temperature is not set by the directors)

(b) $S_{tx} = 1073 - \dfrac{276 \times 36}{10} = 79.4$, $S_{tt} = 7912 - \dfrac{276^2}{10} = 294.4$, $S_{xx} = 154 - \dfrac{36^2}{10} = 24.4$; $r = \dfrac{79.4}{\sqrt{294.4 \times 24.4}} = 0.937$

(c) The directors believe there are *more* failures at higher temperatures. H_0: $\rho = 0$, H_1: $\rho > 0$; critical value ($n = 10$, 1%) 0.7155. $0.937 > 0.7155$ so reject H_0: there is strong evidence of positive correlation

(d) The same, 0.937 (a linear change of units does not alter r)

Question 3

(a) $r = 0.953$

(b) Strong positive correlation: points close to a straight line of positive gradient

(c) H_0: $\rho = 0$; H_1: $\rho > 0$; critical value ($n = 15$, 5%) 0.4409. $0.953 > 0.4409$, so reject H_0: there is evidence that taller students have heavier backpacks (correlation, not causation)

(d) No: r is unchanged by a change of units